FA24-BSE-042 Information Security

1) Malware Attacks:-

Working: It enters system trough malicious files or downloads and execute harmful actions like stealing or damaging data.

Security: Antivirus, keep systems updated, avoid untrusted files.

Enviroment: User devices, enterprise systems, and network with weak endpoint security.

Diagram:

[User]
↓ downloads
[Malicious file]
↓ executes
[System]
↓
[Data theft / Damage]

2) Password Attack:-

Working: The attackers repeatedly tries passwords using brute-force or dictionary attacks until access.

Security: Strong passwords, enable MFA, and account lockout policies.

Enviroment: Login systems, websites, and weak authentication platforms.

Diagram :-

[Attacker]

↓ tries password

repeat ([Login System]

↓ weak password

[Account access Granted]

3) DoS / DDOS :-

Working: The attacker flood the server with excessive requests, making it unavailable to real users.

Security: Apply rate-limiting, firewalls, and traffic filtering / CDN protection.

Environment: Public websites, APIs, and cloud services.

Diagram :-

[Botnet / Attackers]

↓ ↓ ↓ ↓ multiple requests

[server]

↓ overload

[Service Unavailable]

4) Spoofing :-

Working: Attackers pretend to be a trusted source to trick users into sharing data.

Security: Authentication protocols like SPF/DKIM and verifies identities

Environment: Email systems and network communications.

Diagram :-

[Attacker]
↓ creates fake identity.
[Victem]
↓ trusts source
[Sensitive Data shared]

5) MITM :-

The attacker intercepts communication between two parties to read or modify data.

Security : Use HTTPS, encryption and VPNs

Enviroment : Public wifi and insecure network

[User] ⟶ [Server]
↓ Intercepted ↑
[Attacker (MITM)] ⟶ (reads/modifies)

6) Sniffing

The attacker captures network packages to extract sensitive information.

security: Use encryption (HTTPS / VPN) and secure network data.

Enviroment: Open networks and shared LANs.

(User) ⟶ (Network) ⟶ (server)
↓
(Sniffer) captures packets

7) Social Engineering:-

Pschologically manipulation of users to reveal confidential info.

Security: Provide user training and enforce verification procedures.

Enviroment: Organizations with untrained or unaware users.

[Attacker]
↓ manipulates
[User] reveals info → [Confidentiality Compromised]

8) Pharming:-

Redirection of users to fake websites to steal login credentials.

Security: DNS security (DNSSEC) and HTTPS verification.
Enviroment: Browsers and DNS systems.

[User]
↓ enters URL
[DNS Redirect]
↓
[Fake website]
↓
[Credentials stolen]

9) Spam Bombing

Sending a large number of emails to overload or disrupt systems.

Security: Spam filters and rate limiting.

Enviroment: Email servers and inbox systems.

[Attacker] → Many Email → Overload.

10) Backdoor

The attacker intalls hidden access to bypass normal authentication.

Security: Monitor systems, remove unknown accounts, and update software.

Enviroment: Compromised systems and servers.

[Attacker]
↓ installs hidden access
[System]
↓ bypass auth
[Unauthorized Control]

11) Timing Attack:-

Analyzing response time differences to extract sensitive data.

Security: Use constant-time algorithms and secure crypotgraphic design.

Enviroment: Crypographic systems & APIS

[Request]
↓
[Time Measured]
↓
[Secret guessed]

12) Ransomware :-
Encrypt files and demands payment to regain access.

Security: Maintain backups and use endpoint protection
Enviroment : Enterprises and critical systems.

[System]
↓ infected
[files encrypted]
↓
[Ransom Demand]

13) ARP Spoofing

The attacker sends fake ARP messages to redirect network traffic.

Security: secure switches and ARP monitoring tools

Enviroment: LAN

[User] ⟶ [Router]
↓ redirected
[Attacker] ⟶ [Router]
Intercepts traffic

14) Sabotage / Vandalism

The attacker intentionally damages systems or defaces websites.

Security: Apply access control and monitoring

Enviroment: Web servers and internal systems.

[Attacker]
↓ attacks
[System / website]
↓
[Data Damage / Defacement]

15) Human Error

Mistakes by users create vulnerabilities that attackers exploit.

Measures: Provide training and enforce strong policies

Enviroment: Any organization

[User Mistake] ⟶ [Security Weakness]
↓
[System Exploited]

16) Natural Attacks

Natural disasters damage infrastructure and disrupt services.

Security: Use disaster recovery plans and backups.
Enviroment: Data centres and physical infrastructure.

[Natural Disaster]

↓

[Infrastructure damage]

↓

[System Downtime]